ZHENGYUAN JIANG

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EDUCATION

Duke University (Advised by Prof. Neil Gong) **North Carolina, USA**
Ph.D. Student, Electrical and Computer Engineering *2022.9 - 2027 (Expected)*

Duke University **North Carolina, USA**
Master Student, Electrical and Computer Engineering *2022.9 - 2025.9*

University of Science and Technology of China **Hefei, P.R. China**
B.Eng., Information Science and Technology with Honors (top 5%) *2018.9 - 2022.7*

INTERN EXPERIENCE

Alibaba (Ant Group), Research Intern 2025.5 - 2025.8
Collaborator: Zhicong Huang, Cheng Hong

· Conducted research on video watermarking and LLM fingerprinting.
· Two manuscripts under review at top-tier machine learning conferences.

ORNL, Research Intern 2024.5 - 2024.8
Collaborator: Arka Daw, Amir Sadovnik

· Developed methods for detecting AI-generated images.
· One manuscripts under review at CVPR 2026.

SELECTED PUBLICATIONS

Benchmarking Vision-Language Models under Contradictory Virtual Content Attacks in Augmented Reality.
*Yanming Xiu, **Zhengyuan Jiang**, Neil Gong, Maria Gorlatova*. CVPR (Findings) 2026.

Watermark-based Attribution of AI-Generated Content. [Link]
***Zhengyuan Jiang**, Moyang Guo, Yuepeng Hu, Yupu Wang, Neil Gong*. ICLR 2026.

Jailbreaking Safeguarded Text-to-Image Models via Large Language Models. [Link]
***Zhengyuan Jiang**, Yuepeng Hu, Yuchen Yang, Yinzhi Cao, Neil Gong*. EACL (Findings), 2026.

A Transfer Attack to Image Watermarks. [Link]
*Yuepeng Hu, **Zhengyuan Jiang**, Moyang Guo, Neil Gong*. ICLR, 2025.

Certifiably Robust Image Watermark. [Link]
***Zhengyuan Jiang**, Moyang Guo, Yuepeng Hu, Neil Gong*. ECCV, 2024.

AudioMarkBench: Benchmarking Robustness of Audio Watermarking. [Link]
*Hongbin Liu, Moyang Guo, **Zhengyuan Jiang**, Lun Wang, Neil Gong*. NeurIPS, 2024.

Evading Watermark based Detection of AI-Generated Content. [Link]
***Zhengyuan Jiang**, Jinghuai Zhang, Neil Gong*. CCS, 2023.

IPCert: Provably Robust Intellectual Property Protection for Machine Learning. [Link]
***Zhengyuan Jiang**, Minghong Fang, Neil Gong*. ICCV, 2023.

CLMB: deep contrastive learning for robust metagenomic binning. [Link]
*Pengfei Zhang, **Zhengyuan Jiang**, Yixuan Wang, Yu Li*. RECOMB, 2022.

PREPRINT PAPERS

Automatically Refining Coding Rules for AI Coding Agents. [Link]
***Zhengyuan Jiang**, Ruiqi Wang, Yuepeng Hu, Yupu Wang, Neil Gong*. Under review at EMNLP 2026.

EditTrack: Detecting and Attributing AI-assisted Image Editing. [Link]
***Zhengyuan Jiang**, Yuyang Zhang, Moyang Guo, Neil Gong*. Under review at ICML 2026.

Fingerprinting LLMs via Prompt Injection. [Link]
*Yuepeng Hu, **Zhengyuan Jiang**, Mengyuan Li, Osama Ahmed, et al*. Under review at ACL 2026.

SafeText: Safe Text-to-image Models via Aligning the Text Encoder. [Link]
*Yuepeng Hu, **Zhengyuan Jiang**, Neil Gong*. Under review at EMNLP 2026.

SELECTED PROJECTS

EditTrack: Detecting and Attributing AI-assisted Image Editing June 2025 - October 2025

- Proposed EditTrack, the first framework to explicitly address the image-editing detection and attribution problem.
- Introduced a novel re-editing strategy based on four key observations of AI-assisted image-editing.
- Demonstrated that EditTrack consistently achieves accurate detection and attribution, significantly outperforming five state-of-the-art baselines across five image editing models and six datasets.

Fingerprinting LLMs via Prompt Injection June 2025 - October 2025

- Proposed LLMPrint, a novel passive provenance detection framework that constructs fingerprints by exploiting an LLM's inherent vulnerability to prompt injection.
- Developed a unified verification framework that is effective in both the gray-box (access to token-level probabilities) and black-box (only text outputs) settings and demonstrated superior performance over baselines like TRAP and LLMmap.

Jailbreaking Safeguarded Text-to-Image Models via Large Language Models May 2024 - November 2024
Collaborator: Prof. Yinzhi Cao, Johns Hopkins University

- Proposed PromptTune, a query-free jailbreak attack to bypass guardrails of a safeguarded text-to-image model.
- Utilized SFT and DPO to fine-tune a large language model to generate adversarial prompts.
- Demonstrated that three variants of our PromptTune outperform current attacks.

SafeText: Safe Text-to-image Models via Aligning the Text Encoder April 2024 - October 2024

- Proposed SafeText, a novel alignment method for text-to-image models.
- Fine-tuned the text encoder of a text-to-image model to preserve image utility to the greatest extent.
- Demonstrated that SafeText outperforms existing alignment methods for text-to-image models, achieving state-of-the-art performance across three prompt datasets with different models.

TECHNICAL SKILLS

Programming	Python (Advanced), C, Coding Agents
Frameworks	Pytorch, Transformers, Matplotlib, AI Agents
Software&Tools	VSCode, PyCharm, Git, Cursor, Claude Code, Copilot
Soft Skills	Academic Writing & Speaking, Teamwork, Critical Thinking

ADDITIONAL INFORMATION

Research Interests: AI Security, Generative AI, LLM, Image/Video Generation, AI Agent.
Program Committee Service: ICLR, NeurIPS, CVPR, ACL Rolling, AAAI, ect.
Other Interests: Photography, Swimming, Badminton, Road Trip.